CHAPTER 14

EXPLORING OTHER WORLDS AND PROTECTING THIS ONE

> The planets, in their various stages of development, are subjected to the same formative forces that operate on our earth, and have, therefore, the same geologic formation, and probably life, of our own past, and perhaps future; but, further than this, these forces are acting, in some cases, under totally different conditions from those under which they operate on the earth, and hence must evolve forms different from those ever known to man. The value of such material as this to the comparative sciences is too obvious to need discussion.
>
> ROBERT H. GODDARD, NOTEBOOK (1907)

> For the first time in my life, I saw the horizon as a curved line. It was accentuated by a thin seam of dark blue light—our atmosphere. Obviously, this was not the "ocean" of air I had been told it was so many times in my life. I was terrified by its fragile appearance.
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> —ULF MERBOLD, GERMAN SPACE SHUTTLE ASTRONAUT (1988)

When you look down at the Earth from orbital altitudes, you see a lovely, fragile world embedded in black vacuum. But peering at a piece of the Earth through a spacecraft porthole is nothing like the joy of seeing it entire against the backdrop of black, or—better—sweeping across your field of view as you float in space unencumbered by a spacecraft. The first human to have this experience was Alexei Leonov, who on March